EE4-53
EE9-FPN1-02

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

EEE PART IV: MEng and ACGI

HVDC TECHNOLOGY AND CONTROL

Tuesday, 4 May 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions. Questions 1 & 2 should be answered in Booklet A and Question 3 in Booklet B

Questions 1 & 2 carry 25 marks each, Question 3 carries 30 marks

Any special instructions for invigilators and information for candidates are on page 1.

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Answer ALL Questions

Question 1 and 2 carry 25 marks each, Question 3 carries 30 marks

1. a) There are five isolated AC systems (A, B, C, D and E) each with a rated voltage of 400 kV as shown in Figure 1.

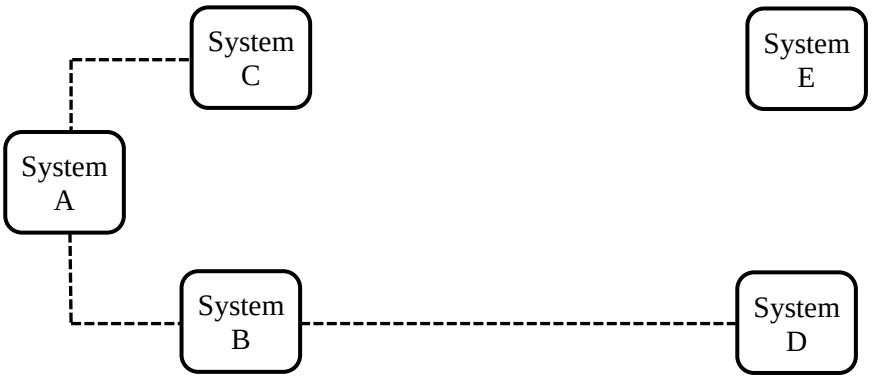


Figure 1.1: Five isolated 400 kV AC systems with planned interconnections shown by dotted lines

The short-circuit level (SCL) at a particular bus within each of these systems where interconnections are planned are provided in Table 1.

System	A	B	C	D	E
SCL (GVA)	20	30	25	50	55

Table 1.1: Short-circuit level at a particular bus for different systems

The plan is to interconnect system B with D via a point-to-point LCC-HVDC link. The reactive power provision at each end of the LCC-HVDC link would be limited to 65% of its rated active power capacity. The minimum effective short circuit ratio (ESCR) is 4.0. Calculate the maximum rated power capacity of the HVDC link connecting B and D under the following situations:

i) Systems A, B are interconnected by a 400 kV AC line with 5 Ω impedance while systems C and E are isolated.

[7]

ii) Systems A and B are interconnected by a 400 kV AC line with 5 Ω impedance while systems A and C are interconnected by a 400 kV AC line with 7.5 Ω impedance.

[5]

b) Consider the standard six-pulse line commutated converter (LCC) configuration with the thyristor valves numbered in the usual order.

i) Assuming ideal thyristor valves, complete Table 1.2 by entering the expressions for the direct voltage (v_d) and voltages across thyristor valve 3 (v_3) and 4 (v_4) when the thyristor valves shown in the first column are conducting. The expressions must be in terms of phase voltage (e.g. e_a, e_b, e_c) or line voltage (e.g. e_{bc}, e_{ca} etc.).

Thyristor valves conducting	v_d	v_3	v_4
$5, 6, 1$			
$6, 1$			
$6, 1, 2$			
$1,2$			

Table 1.2: Direct voltage and thyristor valve voltage for different thyristors in conduction

[1×12]

ii) A commutation failure occurs while thyristor valves $5, 6, 1$ are conducting. Which thyristor valves on the same leg would conduct next creating a short circuit across the converter? Assume there is no commutation failure on the other row.

[1]

2. A 2 GW, ±500 kV point-to-point HVDC link interconnects two separate AC systems. The HVDC link uses a standard 12-pulse converter arrangement for each pole at both ends. At a given time, the AC-side voltage at the rectifier and inverter ends are 402 kV and 393 kV, respectively. The parameters of the converter at each end and the HVDC link are given in Table 2.1:

Minimum firing angle	4°
Minimum extinction advance angle	17°
Resistance of each pole	4.5 Ω
Commutation resistance for each 6-pulse converter	5.0 Ω
Current margin	15%
Turns ratio of converter transformer with low voltage on converter side	0.5

Table 2.1: Parameters of the converter at each end and the HVDC link

a) Under normal conditions, the rectifier end of the HVDC link maintains the rated direct current while the inverter end operates at the minimum extinction advance angle. Calculate the following considering an equivalent monopole configuration:

i) Firing angle (in degrees) at the rectifier.

[5]

ii) Total reactive power consumed by the rectifier.

[2]

iii) Total reactive power consumed by the inverter.

[2]

iv) Commutation overlap angle (in degrees) at the inverter.

[2]

b) If the inverter end AC voltage is reduced by 10%, calculate the following considering an equivalent monopole configuration:

i) Firing angle (in degrees) at the rectifier and extinction advance angle (in degrees) at the inverter.

[4]

ii) Change in total reactive power drawn by the rectifier with respect to the normal condition in part (a).

[3]

iii) Commutation overlap angle (in degrees) at the rectifier.

[2]

iv) Effective turns ratio of the converter transformers at each end needed to maintain the rated direct voltage at the inverter end with minimum reactive power consumption at both ends.

[5]

3. a) Answer the following questions related to the DC-DC converter shown in Figure 3.1. The converter parameters are summarised in Table 3.1:

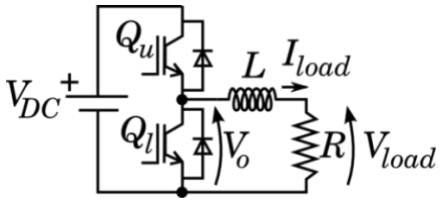


Figure 3.1 – Diagram of a DC-DC converter feeding a resistive load

Parameter	Value	Units	Brief description
V_{DC}	200	V	Rated DC link voltage
R	56	Ω	Load resistor
P_{rated}	400	W	Rated active power delivered to the load
f_{sw}	150	kHz	Rated switching frequency

Table 3.1 – Characteristic parameters of the DC-DC converter

(i) Write the linear ordinary differential equation (ode) that relates $V_o(t)$ and $I_{load}(t)$ and its corresponding transfer function in Laplace variables.

[3]

(ii) Calculate the duty cycle, D, the average of the load current, I_{load}, and the average of the current extracted from the DC bus, I_{DC}, when delivering the rated power to the load.

[3]

(iii) A PI is to be used as a load current controller. The input of the PI will be the error of the current (ie $e=I_{load}^*-I_{load}$) and the output will be the desired average output voltage V_o^* to be given to the PWM. Calculate the gains of the PI, ie K_P and K_I to achieve zero steady state error and a settling time of 1 ms. **Consider L=3 mH.** Justify your answer.

[4]

b) Answer the following questions related to the AC-DC converter shown in Figure 3.2. The converter parameters are summarised in Table 3.2.:

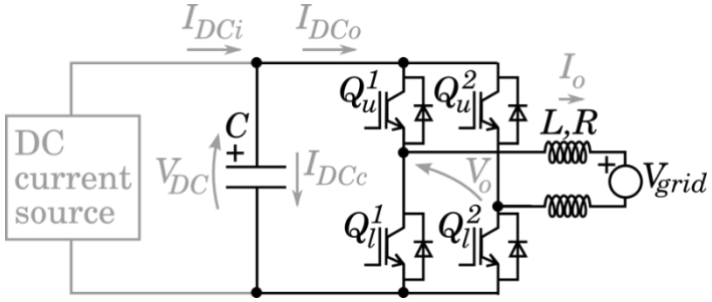


Figure 3.2 – Diagram of a single phase inverter connected to an AC grid

Parameter	Value	Units	Brief description
V_{grid}	120	V rms	Rated AC grid voltage
V_{DC}	300	V	Rated DC link voltage
P_{rated}	1	kW	Rated active power delivered to the grid
f_{rated}	50	Hz	Rated frequency of the AC grid

R	0.1	Ω	Equivalent resistance of bridge plus the filter
L	5	mH	Inductance of the filter

Table 3.2 – Diagram of a single phase inverter connected to an AC grid

(i) Answer the following questions related to the following operating points:

- For an operating point with zero current at the output of the converter. What is the output voltage of the converter, V_o? (use Vgrid phasor as the reference of angle). What is the minimum V_{DC} voltage required to apply V_o?

[3]

- For rated output power ($P_{grid}=P_{rated}$, $Q_{grid}=0$). What is the output current? How much reactive power does the converter bridge have to feed to the output filter? What are the approximate conduction losses?

[2]

- For a given operating point where the 100 Hz component of I_{DCo} is of 3.5 A of amplitude. What is the minimum capacitance, C, the DC bus would need for the 100 Hz ripple of V_{DC} caused by this current to be less than 1V of amplitude?

[2]

(ii) Regarding the design of a DC bus voltage controller for this converter, we know that V_{DC} will have a 100 Hz ripple under normal operation and we can't prevent it. Explain briefly what options we have in order to make the voltage controller ignore the 100 Hz ripple.

[2]

(iii) Regarding the design of the output current controller for this converter, would the steady state error be zero if we chose to use a PI controller? Provide a brief justification.

[3]

c) Consider a basic one-arm modular converter connected between an AC grid and a DC grid represented as equivalent voltage sources as shown in Figure 3.3. Answer the questions below:

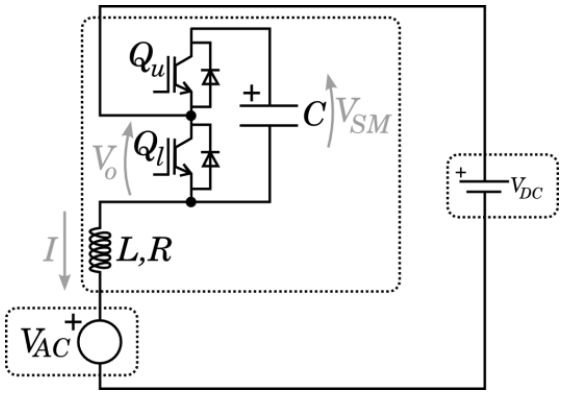


Figure 3.3 – Diagram of a one-arm modular AC/DC converter

Parameter	Value	Units	Brief description
V_{DC}	500	V	Rated DC grid voltage.
V_{AC}	230	V rms	Rated AC grid voltage
f_{rated}	50	Hz	Rated frequency of the AC gr

R	3	mΩ	Equivalent resistance of bridge plus the filter
L	8	mH	Inductance of the filter

Table 3.3 – Diagram of a single phase inverter connected

(i) What is the output voltage of the arm, V_o, when the output current, I, is zero? What is the minimum submodule capacitor voltage, V_{SM}, required to generate this voltage?

[4]

(ii) This converter uses a half-bridge submodule, explain how this affects the minimum V_{DC} required for the converter to operate for a given rated V_{AC}. Could this limitation be overcome by using a different submodule type?

[4]